The weak-coupling gap in a small box reduces to three inequalities through the Schur complement

Conditional scope. H1–H3 are not proved, even at fixed lattice cutoff; formal power counting does not establish them. A lower bound $\Delta \geq \delta_1^{(3)} g^{2/3}(1 - Cg^{2/3})\,\hbar c/L$ for the torus Hamiltonian with a momentum cutoff follows from a proved gap-transfer lemma and three inequalities about polynomial operators on finitely many oscillators. The lemma: if P is an orthogonal projection, $D = \bar{P}H\bar{P} \geq \mu$ on $\operatorname{ran}\bar{P}$, and the Schur-complement error $\eta = \sup_{E \leq a_1} \|(D-E)^{-1/2}\bar{P}HP\|^2$ satisfies $a_1 - \eta < \mu$, then the two lowest eigenvalues of H obey $a_0 - \eta \leq E_0 \leq a_0$ and $E_1 \geq a_1 - \eta$, so $\operatorname{gap}(H) \geq \operatorname{gap}(PHP) - \eta$. Applied with P the projection onto the ground state of the nonzero momentum modes, the three inequalities are: (H1) exciting any nonzero mode costs at least $(2\pi\hbar c/L)(1 - Cg^{2/3})$ in the presence of the constant modes and the couplings; (H2) the coupling between the constant-mode sector and the excited nonzero-mode sector has Schur error $\eta \leq Cg^{4/3}\hbar c/L$, which requires the projection to follow the constant mode (Section 3a) and a gauge-invariant cutoff; (H3) the effective constant-mode Hamiltonian PHP, which is C133's operator plus the nonzero-mode zero-point energy as a function of the constant modes, has gap $\geq \delta_1^{(3)} g^{2/3}(1 - Cg^{2/3})\hbar c/L$. The scaling analysis of the cubic and quartic couplings shows why each inequality is expected with the stated powers, and identifies the smallness parameter as $g^{2/3}$, Lüscher's expansion parameter. The lemma is proved; the three inequalities are the open content of the lower side of the small-volume corner. Constants are kept explicit with the conventions of G07 Proposition 7: g dimensionless, fields of dimension inverse length, every energy a multiple of $\hbar c/L$. Nothing here is promoted.

1. The gap-transfer lemma

Let H be self-adjoint, bounded below, with compact resolvent on a Hilbert space $\mathcal{H}$, and P an orthogonal projection with $P\,D(H) \subset D(H)$; write $\bar{P} = 1 - P$, $A = PHP|_{\operatorname{ran} P}$, $D = \bar{P}H\bar{P}|_{\operatorname{ran}\bar{P}}$, $B = PH\bar{P}$. Let $a_0 \leq a_1 \leq \cdots$ be the eigenvalues of A and $E_0 \leq E_1 \leq \cdots$ those of H.

Lemma 1. Suppose $D \geq \mu$ and, for some $\eta \geq 0$, $\|(D-E)^{-1/2}B^*\|^2 \leq \eta$ for all $E \leq a_1$, with $a_1 < \mu$. Then

$$a_0 - \eta \leq E_0 \leq a_0, \qquad E_1 \geq a_1 - \eta, \qquad E_1 - E_0 \;\geq\; (a_1 - a_0) - \eta.$$

Proof. For $E < \mu$, $D - E$ is invertible and the Schur complement of $H - E$ with respect to the decomposition $\operatorname{ran} P \oplus \operatorname{ran}\bar{P}$ is $F(E) = A - E - B(D-E)^{-1}B^*$ on $\operatorname{ran} P$. The block factorization

$$H - E = \begin{pmatrix} 1 & B(D-E)^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} F(E) & 0 \\ 0 & D-E \end{pmatrix} \begin{pmatrix} 1 & 0 \\ (D-E)^{-1}B^* & 1 \end{pmatrix}$$

is a congruence, so by Sylvester's law of inertia (the number of negative eigenvalues, finite here because H has compact resolvent and $E < \mu \leq \inf D$) the number of eigenvalues of H below E equals the number of negative eigenvalues of $F(E)$, since $D - E > 0$ contributes none. The hypothesis gives $0 \leq B(D-E)^{-1}B^* \leq \eta$, hence $A - E - \eta \leq F(E) \leq A - E$ and, by min-max, the number of negative eigenvalues of $F(E)$ lies between $\#\{k : a_k < E\}$ and $\#\{k : a_k < E + \eta\}$. Taking E slightly above a_0 shows $E_0 \leq a_0$; taking $E = a_0 - \eta$ shows no eigenvalue of H lies below $a_0 - \eta$; taking $E = a_1 - \eta$ shows at most one eigenvalue of H lies below $a_1 - \eta$, so $E_1 \geq a_1 - \eta$. □

The lemma transfers a gap from the compressed operator PHP to H at the price η, which is the size of the coupling between $\operatorname{ran} P$ and its complement measured against the energy $D - E$ needed to leave $\operatorname{ran} P$. Everything below is the choice of P and the estimation of $a_1 - a_0$, μ and η.

2. The cutoff torus Hamiltonian and the projection

On T^3_L in temporal gauge, with $SU(2)$ for definiteness, momentum cutoff $|k| \le \Lambda$ and the convention of G07 Proposition 7 ($S = -\frac{1}{4g^2}\int F^a F^a$), decompose $A_i(x) = a_i + \tilde{A}_i(x)$ into the constant mode and the nonzero modes $\tilde{A}_i(x) = L^{-3}\sum_{0<|k|\le\Lambda} \hat{A}_i(k) e^{ik\cdot x}$, with conjugate momenta p_i and $\tilde{\Pi}_i$. Then

$$H = H_0 + H_\perp + W,$$

$$H_0 = \frac{g^2 c}{2\hbar L^3}\sum_i |\vec{p}_i|^2 + \frac{\hbar c L^3}{2g^2}\sum_{i<j}|\vec{a}_i \times \vec{a}_j|^2, \qquad H_\perp = \sum_{0<|k|\le\Lambda}\Big[\frac{g^2 c}{2\hbar}|\hat{\Pi}(k)|^2 + \frac{\hbar c}{2g^2}|k\times\hat{A}(k)|^2\Big]$$

(with $[a^a_i, p^b_j] = i\hbar\delta_{ij}\delta^{ab}$ and the corresponding normalization of $\hat{\Pi}$, so that the free part has energies $\hbar c|k|$), and W collects every term of $\frac{\hbar c}{2g^2}\int |F_{ij}|^2$ that is neither purely constant-mode nor quadratic in $\tilde{A}$ at $a = 0$:

$$W = W_2 + W_3 + W_4, \quad W_2 = \frac{\hbar c}{g^2}\int (\partial_i\tilde{A}_j - \partial_j\tilde{A}_i)\cdot([a_i,\tilde{A}_j] - [a_j,\tilde{A}_i]) + \frac{\hbar c}{2g^2}\int |[a_i,\tilde{A}_j] - [a_j,\tilde{A}_i]|^2,$$

$$W_3 = \frac{\hbar c}{g^2}\int \bar{F}_{ij}\cdot[\tilde{A}_i,\tilde{A}_j], \qquad \bar{F}_{ij} = \partial_i\tilde{A}_j - \partial_j\tilde{A}_i + [a_i,\tilde{A}_j] - [a_j,\tilde{A}_i] + [a_i,a_j],$$

$$W_4 = \frac{\hbar c}{2g^2}\int |[\tilde{A}_i,\tilde{A}_j]|^2,$$

where the purely constant part $\frac{\hbar c L^3}{2g^2}|[a_i,a_j]|^2$ of $\frac{\hbar c}{2g^2}\int|F|^2$ is in H_0 and W_4 is quartic in the nonzero modes. The Gauss constraint is imposed on states; the constant gauge transformations act on a_i by the adjoint action and on the nonzero modes covariantly, and the nonconstant ones are fixed by keeping the transverse nonzero modes, the standard small-volume gauge choice.

Let $\Omega_\perp$ be the ground state of $H_\perp$, the Gaussian vacuum of the free transverse modes, with energy $E_\perp = \hbar c\sum_k |k|$ (a constant that cancels from gaps) and first excitation energy $\mu_\perp = 2\pi\hbar c/L$. Set

$$P = 1_{\text{const}} \otimes |\Omega_\perp\rangle\langle\Omega_\perp|.$$

P commutes with the constant gauge transformations because $\Omega_\perp$ is invariant under them, so Lemma 1 can be applied inside the physical sector. The compression is

$$A = PHP = H_0 + \langle\Omega_\perp, W\Omega_\perp\rangle(a) + E_\perp,$$

where $\langle\Omega_\perp, W_2\Omega_\perp\rangle(a)$ is a function of the constant modes only (the Gaussian expectation of a quadratic form in $\tilde{A}$ with a-dependent coefficients), $\langle W_3\rangle$ vanishes (odd in $\tilde{A}$), and $\langle W_4\rangle$ is a constant. Thus A is C133's operator plus a potential $U(a) = \langle W_2\rangle(a)$, which is the leading change of the nonzero-mode zero-point energy with the constant background.

3. The three inequalities and their expected sizes

Rescale $\vec{a}_i = g^{2/3}L^{-1}\vec{\xi}_i$ and $\hat{A}(k) = g\,\hat{A}'(k)$ so that $H_0 = g^{2/3}(\hbar c/L)\,h_3(\xi)$ with h_3 the unit operator of C133 and $H_\perp$ is g-independent in the primed variables. Then, counting powers of g at fixed L and at typical values $|\xi| = O(1)$, $|\hat{A}'| = O(1)$:

term	order in units of $\hbar c/L$	role
H_0	$g^{2/3}$	the gap to be transferred
$H_\perp$ excitations	$\ge 2\pi$	the energy μ of leaving ran P
W_2, linear in a	$g^{2/3}$	enters $U(a)$ and B
W_2, quadratic in a	$g^{4/3}$	enters $U(a)$

term	order in units of $\hbar c/L$	role
W_3, term with $\partial\tilde{A}$	g	enters B
W_3, terms with a	$g^{5/3}$, $g^{7/3}$	enters B
W_4	g^2	enters A and D

With these sizes the three inequalities that Lemma 1 needs are:

(H1) $D = \bar{P}H\bar{P} \;\geq\; E_\perp + \frac{2\pi\hbar c}{L}(1 - C_1 g^{2/3})$ on $>$ ran $\bar{P}$: any state with an excited nonzero mode costs $>$ at least one quantum $2\pi\hbar c/L$ up to the shift produced by W in a background $|a| \sim g^{2/3}/L$.

(H2) $\eta = \sup_{E \leq a_1} \|(D-E)^{-1/2}\bar{P}WP\|^2 \;\leq\; C_2\, g^{4/3}\hbar c/L$. $>$ The leading contribution is the W_2 term linear in a (order $> g^{2/3}\hbar c/L$, creating one or two nonzero quanta) squared and $>$ divided by $2\pi\hbar c/L$; the W_3 term of order $g\hbar c/L >$ contributes $g^2\hbar c/L$.

(H3) $\mathrm{gap}(A) = \mathrm{gap}\,(H_0 + U(a)) \;\geq\; \delta_1^{(3)} g^{2/3}(1 - C_3 g^{2/3})\,\hbar c/L$. $> U(a)$ is even in a (charge conjugation), its linear term vanishes, $>$ and its quadratic term is of order $g^{4/3}\hbar c/L$, a relatively bounded perturbation of h_3 in the rescaled variables.

Given (H1)–(H3), Lemma 1 with $\mu = E_\perp + \frac{2\pi\hbar c}{L}(1 - C_1 g^{2/3})$ and $a_1 - \eta < \mu$ (true for small g since $a_1 - a_0 = O(g^{2/3}\hbar c/L)$ and $a_0 - E_\perp = O(g^{2/3}\hbar c/L)$) yields

$$\Delta \;\geq\; \delta_1^{(3)}\, g^{2/3}\, \frac{\hbar c}{L}\,(1 - C g^{2/3}).$$

This conditional bound would give the lower side of the small-volume corner S, comparable with the upper side obtained in the Polyakov-average note up to pure numbers, and it is the leading term of Lüscher's expansion (abstract, B78) as a two-sided bound.

3a. Correction: the projection must follow the constant mode, and the cutoff must be gauge invariant

The counting in the table is right for the sizes of the operators but (H2) fails for the projection P of Section 2 when the cutoff is a sharp momentum cutoff. The term of W_2 linear in a creates pairs of nonzero quanta $(k, -k)$ from $\Omega_\perp$ with amplitude of order $g^{2/3}|\xi|/L$ for every k, so $\|\bar{P}W_2 P\psi\|^2 \sim (g^{4/3}/L^2)(\Lambda L)^3|\xi|^2$ and, after dividing by the pair energy $2|k|$,

$$\eta \;\sim\; \frac{g^{4/3}\hbar c}{L^2} \sum_{0<|k|\leq\Lambda} \frac{|\xi|^2}{2|k|} \;\sim\; \frac{g^{4/3}\hbar c}{L}\,(\Lambda L)^2\,|\xi|^2,$$

quadratically divergent in the cutoff. This is the second-order energy shift $-c\,|a|^2\Lambda^2 L^3$ of the constant mode. A quadratic term in a is allowed by the constant gauge transformations, which act by conjugation and leave $\mathrm{tr}\,a_i^2$ invariant, and the exact one-loop potential of the constant modes on the torus does contain one: the sum $\frac{1}{2}\sum_k[\omega_k(a) - \omega_k(0)]$ is a periodic function of the holonomy angles La_i (shift of the summation variable), so its small-a expansion is $\hbar c\,[c_2(La)^2/L + O((La)^4/L)]$ with a finite pure number c_2, the Casimir-type Weiss potential, of order $g^{4/3}\hbar c/L$ in the rescaled variables and cutoff independent. The quadratic Λ^2 divergence above is an artefact of the sharp momentum cutoff, which breaks the periodicity in La; a periodic (lattice) cutoff, or any regularization compatible with the large gauge transformations, gives the finite coefficient. Two changes make the reduction usable.

Fibered projection. $H_\perp + W_2$ is quadratic in $\tilde{A}$ at fixed a, so it has, for $|a|$ small against $2\pi/L$, a Gaussian ground state $\Omega_\perp(a)$ with frequencies $\omega_k(a)$, the spectrum of the covariant transverse

curl in the constant background (for non-commuting a_i the background carries a constant field strength and the lowest shifted frequency is the Nielsen–Olesen mode, $\omega^2 = (2\pi/L)^2 - c\,|[a,a]|$, still positive when $g^{4/3} \ll (2\pi)^2$). Take

$$P = \int^{\oplus} da\, |\Omega_\perp(a)\rangle\langle\Omega_\perp(a)|,$$

the Born–Oppenheimer projection. Then W_2 is absorbed exactly, the compression is

$$A = PHP = H_0 + U(a) + (\text{non-adiabatic terms}), \qquad U(a) = \tfrac{1}{2}\sum_k \left[\omega_k(a) - \omega_k(0)\right] + \text{const},$$

and $U(a)$ is the one-loop effective potential of the constant modes on the torus: even in a, with the finite quadratic Casimir term $c_2(La)^2/L$, a finite quartic term, and a quartic term $\propto |[a,a]|^2 \log(\Lambda L)$ that renormalizes $1/g^2$. In rescaled variables the finite part of U is $O(g^{4/3}/L)$ and the logarithm is $O(g^{4/3} \cdot g^2 \log \Lambda L\,/L)$ relative to H_0's $g^{2/3}/L$: (H3) remains an inequality to prove, even if g is interpreted as a renormalized coupling at scale L. The assertion that a periodic cutoff makes the a-independent projection work is unproved: cancellation in a scalar energy sum does not supply the relative operator bound; the fibered projection is cleaner because it removes W_2 from the error altogether. The non-adiabatic terms come from p_i acting on $\Omega_\perp(a)$ (the Berry connection of the fibered vacuum) and are of relative order $g^{4/3}$.

Schur error. With the fibered P, $\bar{P}WP$ contains only W_3, W_4 and the non-adiabatic terms. W_3 creates three quanta with amplitude of order g/L per mode triple, and its second-order contribution is the a-dependent part of a two-loop vacuum energy, whose divergent part is a-independent and cancels from the gap; the finite a-dependent part is of order $g^2\hbar c/L$. (H2) is therefore expected in the form $\eta \le C_2 g^{4/3}(1 + g^{2/3}\log \Lambda L)\,\hbar c/L$, with a gauge-invariant cutoff. Neither the constants nor the relative inequality have been established; a logarithmic estimate alone does not identify the running of the coupling or complete the reduction at fixed cutoff.

Relative form of the lemma. The constant-mode coordinate ξ is unbounded and the pair amplitudes grow with $|\xi|$, so η cannot be a uniform operator bound; the usable hypothesis is relative to A: $B(D-E)^{-1}B^* \le \epsilon(A-a_0)+\eta$ with $0 \le \epsilon < 1$. Then $F(E) \ge (1-\epsilon)(A-a_0)+a_0-E-\eta$, the count of negative eigenvalues of $F(E)$ is at most $\#\{k : a_0+(1-\epsilon)(a_k-a_0)-\eta < E\}$, and the same argument gives $E_0 \ge a_0 - \eta$, $E_1 \ge a_0 + (1-\epsilon)(a_1-a_0) - \eta$, hence $\mathrm{gap}(H) \ge (1-\epsilon)(a_1-a_0) - \eta$. The confining potential of h_3 controls $\langle|\xi|^2\rangle$ in terms of $A - a_0$, which is what makes ϵ small, of order $g^{2/3}$.

The corrected list is: (H1) as before for $H_\perp(a)$; (H2) in the relative form for the fibered Schur error; (H3) for $h_3 + U$ with U the finite one-loop torus potential including its Casimir quadratic term.

4. What the reduction shows

The smallness parameter is $g^{2/3}$ in every entry of the table, which is the reason the small-volume expansion is a series in $g^{2/3}$. The mechanism transferred is the zero-point confinement of C133: A is that operator plus a small even potential, and the nonzero modes enter only through the price η of leaving their vacuum. The reduction also suggests a possible crossover when the assumed H1 lower bound ceases to separate scales, for example when $\Delta_0(L) \sim 2\pi\hbar c/L$, which is the crossover $z \simeq 2$ of Lüscher–Münster in the variable $z = \Delta L$. Beyond it, leaving the nonzero-mode vacuum is no longer expensive compared with the constant-mode gap, no projection of the form P separates scales, and the argument gives nothing, which is the statement of T2′'s difficulty in this language.

4a. Schur error and the ultraviolet obstruction

The following refines Lemma 1 and retains the earlier valley calculations as formal power counting, not established inequalities. In particular, the previously asserted fixed-lattice theorem was

not proved: H1–H3, the domain and uniform constants of the fibered projection, and the renormalized matching remain open. A logarithm of the right order alone does not prove that the Berry term is absorbed into a running coupling. The earlier cubic slope count used $(\Lambda L)^2$; its displayed six-dimensional integrand instead has superficial degree three, so the formal scaling is $g^2(\Lambda L)^3$ and its smallness would require $g(\Lambda L)^{3/2} \ll 1$. The Berry coefficient below also needs its charge and polarization multiplicities rechecked; it is not a certified constant. See the corpus audit §3. The estimates refer to a specified $SU(2)$ abelian-valley calculation with cutoff Λ, not an $SU(3)$ or continuum gap theorem.

4a.1. The refined transfer lemma with a constant shift

Lemma 1 above bounds the Schur term $S(E) = B(D-E)^{-1}B^*$ by a constant. When $S(E)$ contains a large part proportional to the identity, that part shifts all eigenvalues of the compression equally and must not be charged to the gap.

Lemma 1′. In the setting of Lemma 1 suppose that for $E \in I = [a_0 - \eta_1 - \bar\eta_0,\, a_1]$,

$$S(E) = \eta_0(E)\,1 + R(E), \qquad 0 \le R(E) \le \epsilon(A - a_0) + \eta_1,$$

with η_0 a nonnegative nondecreasing function, $\bar\eta_0 = \sup_I \eta_0$, $0 \le \epsilon < 1$, and $\eta_0(E)$ differentiable with $0 \le \eta_0'(E) \le \theta < 1$ on I. Then

$$E_1 - E_0 \;\ge\; \frac{(1-\epsilon)(a_1 - a_0) - 2\eta_1}{1+\theta}.$$

Proof. $F(E) = A - E - \eta_0(E) - R(E)$. The count of negative eigenvalues of $F(E)$ lies between the counts of $A - E - \eta_0(E) - \epsilon(A - a_0) - \eta_1$ and $A - E - \eta_0(E)$. Write $\tilde E = E + \eta_0(E)$, which is increasing in E with $d\tilde E/dE \le 1 + \theta$. The bounds show that E_0 is the solution of $\tilde E = e_0$ with $e_0 \in [a_0 - \eta_1, a_0]$ and E_1 the solution of $\tilde E = e_1$ with $e_1 \ge a_0 + (1-\epsilon)(a_1 - a_0) - \eta_1$; since $\tilde E$ has slope at most $1 + \theta$, $E_1 - E_0 \ge (e_1 - e_0)/(1+\theta)$. □

The constant shift costs nothing; its energy dependence costs the factor $1/(1+\theta)$, and θ is the quantity that the cubic coupling makes large.

4a.2. The fibered vacuum and its a-derivative on the abelian valley

On the abelian valley of the valley note, $a = \alpha T^3$, the charged modes at momentum k have energies $\hbar\omega_k^{(\pm)} = \hbar c|k \pm a|$ and the fibered vacuum is the product of Gaussians $\Omega_\perp(a) = \prod_k \Omega_k(a)$ with $\Omega_k \propto \exp[-\omega_k(a)|q_k|^2/(2\hbar \cdot \mathrm{norm})]$ in the real mode coordinates q_k of each polarization and charge. For a single oscillator of frequency ω the derivative of the normalized ground state with respect to ω is $\partial_\omega \Omega = -\frac{1}{2\sqrt{2}\,\omega}\,\Omega_2$, with Ω_2 the normalized second excited state, so

$$\bar P\,\partial_{a_i}\Omega_\perp(a) = -\sum_{k,\pm,\mathrm{pol}} \frac{\partial_{a_i}\omega_k^{(\pm)}}{2\sqrt{2}\,\omega_k^{(\pm)}}\;\Omega_{k,2} \otimes \prod_{k' \ne k} \Omega_{k'}, \qquad \frac{\partial_{a_i}\omega_k^{(\pm)}}{\omega_k^{(\pm)}} = \pm\frac{(k \pm a)_i}{|k \pm a|^2}.$$

Its norm squared is $\sum_{k,\pm,\mathrm{pol}} \frac{(k\pm a)_i^2}{8|k\pm a|^4} \simeq \frac{2\cdot 2}{8}\sum_k \frac{\hat k_i^2}{|k|^2} = \frac{1}{2}\left(\frac{L}{2\pi}\right)^3 \frac{4\pi}{3}\Lambda\,[1 + O(a/\Lambda)] = \frac{L^3\Lambda}{12\pi^2}$, linearly divergent: the vacua at different a are not close in norm as the cutoff is removed, which is the first sign that the fiber is ultraviolet-sensitive. What enters the Schur error is the same sum with the energy denominator, which converges better.

4a.3. The Berry term

The constant-mode kinetic operator $\frac{g^2 c}{2\hbar L^3}\sum_i |\vec p_i|^2$ with $p_i = -i\hbar\partial_{a_i}$ acts on $\psi(a)\Omega_\perp(a)$ as $\frac{g^2 c}{2\hbar L^3}\big[(-\hbar^2\Delta_a\psi)\Omega_\perp - 2\hbar^2(\partial_{a_i}\psi)(\partial_{a_i}\Omega_\perp) - \hbar^2\psi\,\Delta_a\Omega_\perp\big]$. The $\bar P$-component of the middle term is

the leading non-adiabatic coupling,

$$B_{\text{Berry}}\,\psi = -\frac{g^2 c\hbar}{L^3}\sum_i (\partial_{a_i}\psi)\,\bar{P}\,\partial_{a_i}\Omega_\perp .$$

Each two-quantum state $\Omega_{k,2}$ costs $D-E \geq 2\hbar\omega_k$ above the fiber, so

$$\|(D-E)^{-1/2}B_{\text{Berry}}\psi\|^2 \simeq \Big(\frac{g^2 c\hbar}{L^3}\Big)^2 \sum_i \sum_{k,\pm,\text{pol}} \frac{(k_i)^2}{8|k|^4\cdot 2\hbar c|k|}\,|\partial_{a_i}\psi|^2 = \frac{g^4 c\hbar}{16L^6}\sum_i \Big(\sum_k \frac{\hat{k}_i^2}{|k|^3}\Big)|\partial_{a_i}\psi|^2,$$

with $\sum_k \hat{k}_i^2|k|^{-3} \simeq \frac{1}{3}\big(\frac{L}{2\pi}\big)^3 4\pi \int_{2\pi/L}^{\Lambda}\frac{dk}{k} = \frac{L^3}{6\pi^2}\log\frac{\Lambda L}{2\pi}$. In the rescaled variable $a = g^{2/3}\xi/L$, $\partial_a = (L/g^{2/3})\partial_\xi$, and

$$\eta_{\text{Berry}} \;\simeq\; \frac{g^4 c\hbar}{16L^6}\cdot\frac{L^3}{6\pi^2}\log(\Lambda L)\cdot\frac{L^2}{g^{4/3}}\,\langle|\partial_\xi\psi|^2\rangle = \frac{g^{8/3}}{96\pi^2}\,\log(\Lambda L)\,\langle|\partial_\xi\psi|^2\rangle\,\frac{\hbar c}{L}.$$

Since $\langle|\partial_\xi\psi|^2\rangle$ is controlled by the kinetic part of h_3, this is a relative bound of the form $R \leq \epsilon(A-a_0)+\eta_1$ with $\epsilon \sim g^2\log(\Lambda L)/(96\pi^2)$. In this formal estimate the logarithm is the ultraviolet signature of vacuum polarization by the constant background; the one-loop coefficient of the running coupling for $SU(2)$ is $b_0 = 11/(24\pi^2)$, and the Berry term is one of the contributions that together must reproduce it, which is the consistency check the renormalized expansion has to pass. Its relative size is formally $g^2\log(\Lambda L)$; whether the complete gauge-compatible calculation absorbs it into a renormalized coupling remains to be shown.

4a.4. The cubic coupling: constant shift and slope

$W_3 = \frac{\hbar c}{g^2}\int \bar{F}_{ij}\cdot[\tilde{A}_i,\tilde{A}_j]$ with $\tilde{A} = g\tilde{A}'$ creates three quanta from the fiber vacuum. With the mode normalization $\tilde{A}' \sim \sqrt{\hbar/(2\omega L^3)}(b+b^\dagger)$ the amplitude for (k,k',k''), $k+k'+k''=0$, is of order $g\hbar c\,|k|\,(\hbar/(2\hbar c))^{3/2}L^{-3/2}(|k||k'||k''|)^{-1/2}$ times colour factors, and the second-order shift is

$$\eta_0 \;\sim\; g^2\hbar c\sum_{k,k'}\frac{|k|^2/L^3}{|k||k'||k''|\,(|k|+|k'|+|k''|)} \;\sim\; g^2\hbar c\,L^3\Lambda^4\cdot\text{const} = g^2\,\frac{\hbar c}{L}\,(\Lambda L)^4\cdot\text{const},$$

the two-loop vacuum energy of the box: extensive and quartically divergent, and independent of a at leading order. By Lemma 1′ it costs nothing as a shift. Its energy derivative is

$$\theta = \eta_0'(E) \;\sim\; g^2\sum_{k,k'}\frac{|k|^2/L^3}{|k||k'||k''|\,(|k|+|k'|+|k''|)^2}\cdot\frac{1}{\hbar c}\cdot\hbar c \;\sim\; g^2(\Lambda L)^3\cdot\text{const},$$

and Lemma 1′ needs $\theta<1$: this crude undressed estimate would require $g^2(\Lambda L)^3 \ll 1$. On a lattice $\Lambda L = \pi N_s$, so the formal condition is $g \ll N_s^{-3/2}$. The a-dependent part of the W_3 shift requires separate gauge-compatible renormalization; its formal expected order does not establish a bound.

5. Consequence for STATE

The weak-coupling lower bound is now three inequalities on explicit polynomial operators, with a proved transfer lemma. The next step on the lower side is the fibered (H2): the non-adiabatic terms and the W_3 Schur error against the nonzero-mode gap, on a lattice cutoff, with the C133 ground state as weight; (H1) and (H3) are relatively bounded perturbation statements, (H3) requiring the finite one-loop torus potential. Their proof would close the small-volume corner S from both sides and make a conditional small-volume result; controlling the crossover, physical spectral transfer and the continuum remain separate obligations.